Design and Implementation of a Network Intrusion Detection System Using Machine Learning

Student: Lioto Samuel Simon | **Supervisor:** Dr. Felix Gonda

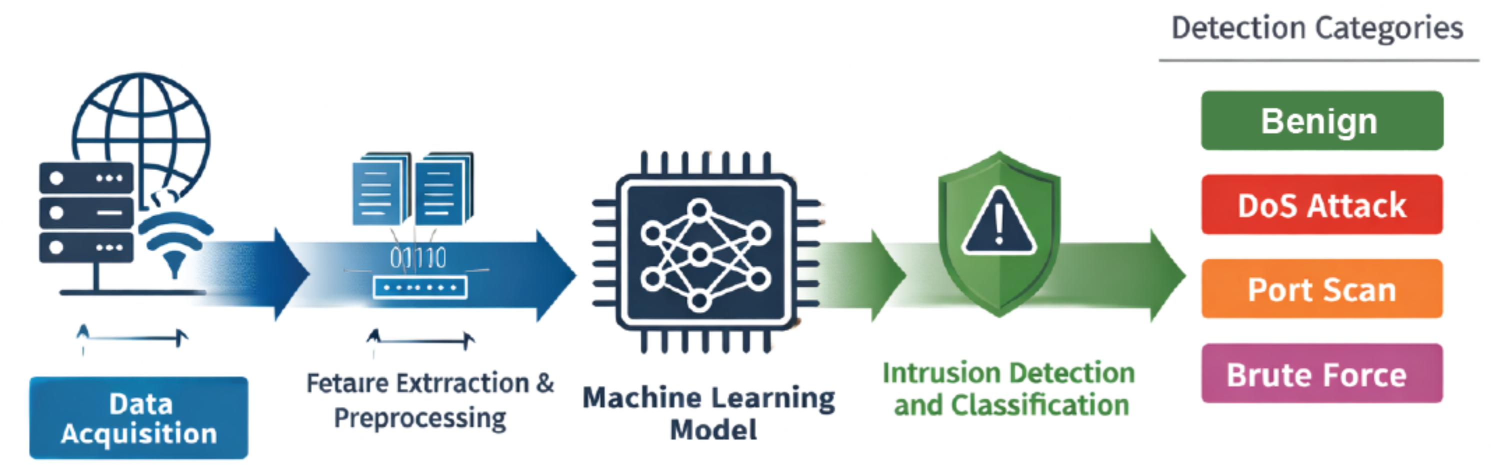
Problem & Motivation

- **Problem Context:** Traditional signature-based IDS struggle to detect evolving cyberattacks.
- **Research Gap:** Limited adaptive detection of unknown attacks with high accuracy.
- **Objective:** Design a CNN-based NIDS to classify normal network traffic and malicious network traffic.

Methodology

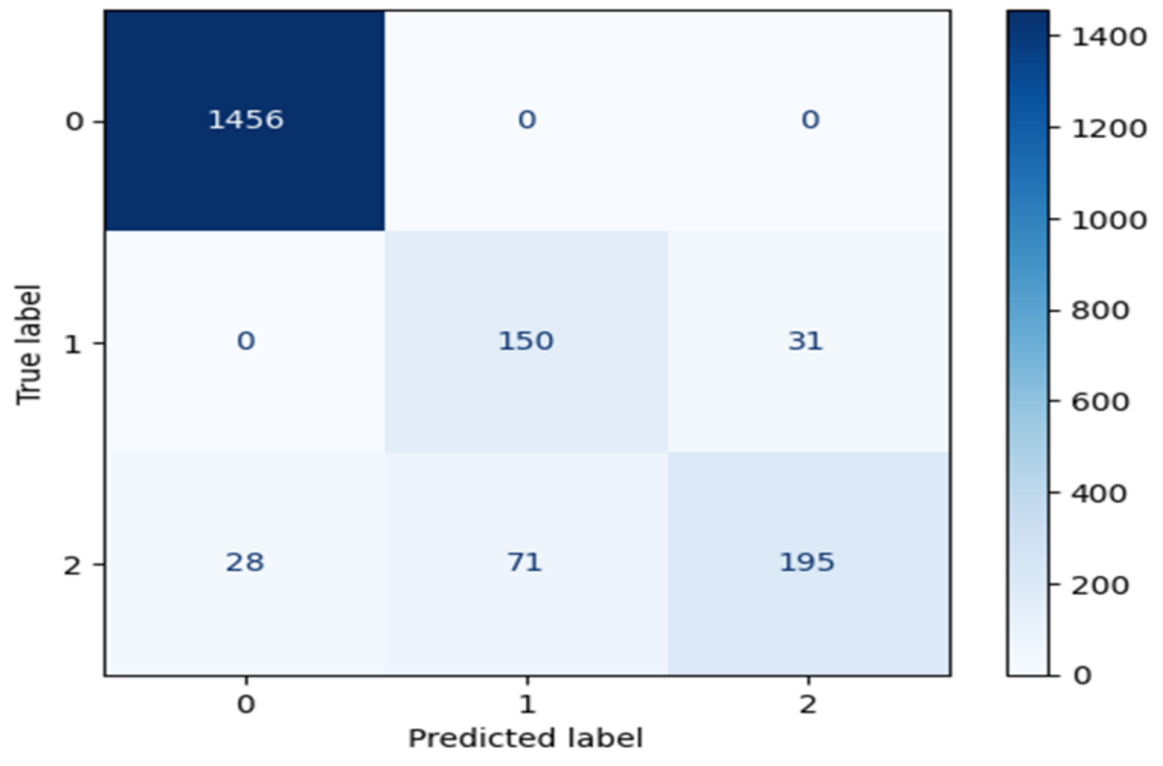
System Architecture

*Our system utilizes a Multilayer Perceptron (**MLP**) network for heart attack risk classification. This architecture is well-suited for processing tabular numeric data and mapping multi-variable patient metrics into discrete risk categories.*



Evaluation & Key Results

*Our MLP achieved **93% test accuracy**. Confusion matrix is show below:*



Dataset

Our data is sourced from the CICIDS2017 networking dat which includes more than 80 numeric and categorical features collected over 5 days of simulated traffic.

Flow Duration	TotFwdPkts	FlowIATMean	…	Label
12342	5	3500	…	BENIGN
250	200	50	…	DoS Hulk
500	1	200	…	PortScan
90	150	20		DDoS

Significance & Conclusion

- **Contribution:** Developed an intelligent CNN-based intrusion detection framework enhanced with SMOTE.
- **Impact:** Improves detection or DoS/DDoS, Brute Force and PortScan attacks for modern networks.
- **Future Work:** Deploy on live network traffic, optimize for real-time performance, and evaluate additional attack classes.